

Here's a list of 100 commonly asked questions in digital electronics along with their answers:

1. What is Digital Electronics?

Answer: Digital electronics deals with circuits that work with discrete values, usually represented as 0s and 1s (binary digits). It involves the use of logic gates to process binary information.

2. What is a logic gate?

Answer: A logic gate is a basic building block of digital circuits that performs a logical operation on one or more binary inputs and produces a single output.

3. What are the basic logic gates?

Answer: AND, OR, NOT, NAND, NOR, XOR, and XNOR.

4. What is a NOT gate?

Answer: A NOT gate inverts the input; if the input is 1, the output is 0, and if the input is 0, the output is 1.

5. What is an AND gate?

Answer: An AND gate outputs 1 only if both of its inputs are 1; otherwise, it outputs 0.

6. What is an OR gate?

Answer: An OR gate outputs 1 if at least one of its inputs is 1; if both inputs are 0, the output is 0.

7. What is a NAND gate?

Answer: A NAND gate is the opposite of an AND gate. It outputs 0 only if both inputs are 1, otherwise, it outputs 1.

8. What is a NOR gate?

Answer: A NOR gate is the opposite of an OR gate. It outputs 1 only if both inputs are 0, otherwise, it outputs 0.

9. What is an XOR gate?

Answer: An XOR (exclusive OR) gate outputs 1 if the inputs are different, and 0 if they are the same.

10. What is an XNOR gate?

Answer: An XNOR gate is the opposite of an XOR gate. It outputs 1 if the inputs are the same, and 0 if they are different.

11. What is Boolean Algebra?

Answer: Boolean algebra is a branch of mathematics that deals with variables and operations using binary values (0 and 1) and logical operators.

12. What is the truth table of an AND gate?

Answer:

13. What is the truth table of an OR gate?

Answer:

14. What is the truth table of a NOT gate?

Answer:

15. What is the truth table of a NAND gate?

Answer:

16. What is the truth table of a NOR gate?

Answer:

17. What is the truth table of an XOR gate?

Answer:

18. What is the truth table of an XNOR gate?

Answer:

19. What is a Half Adder?

Answer: A Half Adder is a combinational circuit that adds two binary digits and outputs a sum and a carry.

20. What is a Full Adder?

Answer: A Full Adder is a combinational circuit that adds three binary digits (two significant bits and a carry input) and outputs a sum and a carry.

21. What is a Half Subtractor?

Answer: A Half Subtractor is a combinational circuit that subtracts two binary digits and provides a difference and borrow.

22. What is a Full Subtractor?

Answer: A Full Subtractor is a combinational circuit that subtracts three binary digits and provides a difference and borrow.

23. What is a Multiplexer (MUX)?

Answer: A Multiplexer is a digital switch that selects one of many inputs and forwards the selected input to a single output.

24. What is a Demultiplexer (DEMUX)?

Answer: A Demultiplexer is a digital switch that takes a single input and channels it to one of many outputs based on a control signal.

25. What is a Flip-Flop?

Answer: A Flip-Flop is a bistable circuit that has two states and can store one bit of data.

26. What is the difference between a latch and a flip-flop?

Answer: A latch is level-sensitive, while a flip-flop is edge-triggered, meaning flip-flops change states only on a clock signal edge.

27. What is a D Flip-Flop?

Answer: A D Flip-Flop is a type of flip-flop that captures the input value at the moment of a clock edge and holds that value until the next clock edge.

28. What is a JK Flip-Flop?

Answer: A JK Flip-Flop is a type of flip-flop that has two inputs, J and K, and can toggle its output, set, or reset based on the inputs and clock signal.

29. What is an SR Flip-Flop?

Answer: An SR (Set-Reset) Flip-Flop is a bistable device with two inputs, Set (S) and Reset (R), and is used to store a binary value.

30. What is a T Flip-Flop?

Answer: A T Flip-Flop is a type of flip-flop where the output toggles between 1 and 0 with each clock pulse when the T input is 1.

31. What is a Counter?

Answer: A Counter is a sequential circuit that counts the number of pulses of a clock signal and outputs a binary number representing the count.

32. What is a Register?

Answer: A Register is a group of flip-flops used to store multiple bits of data.

33. What is a Shift Register?

Answer: A Shift Register is a type of register that shifts its stored data to the left or right with each clock pulse.

34. What is the difference between synchronous and asynchronous circuits?

Answer: Synchronous circuits operate based on a clock signal, while asynchronous circuits operate without a clock and rely on the input signal.

35. What is the purpose of a Clock Signal in digital circuits?

Answer: A Clock signal is used to synchronize the operation of flip-flops, counters, and other sequential circuits.

36. What is the principle of operation of a NAND gate?

Answer: A NAND gate outputs 0 only when both of its inputs are 1. Otherwise, it outputs 1.

37. What is the principle of operation of an OR gate?

Answer: An OR gate outputs 1 if at least one of its inputs is 1; otherwise, it outputs 0.

38. What is the significance of De Morgan's Theorem?

Answer: De Morgan's Theorem simplifies the process of working with inverted logic operations. It states that the NOT of an AND operation is the same as the OR of the NOTs and vice versa.

39. What are the types of flip-flops used in digital electronics?

Answer: SR Flip-Flop, D Flip-Flop, JK Flip-Flop, T Flip-Flop.

40. What is a Gray Code?

Answer: A Gray Code is a binary numeral system where two successive values differ in only one bit, which is useful in minimizing errors in digital communications and electronics.

41. What is the purpose of an adder circuit?

Answer: An adder circuit performs the addition of binary numbers and outputs a sum and a carry.

42. What is a BCD (Binary-Coded Decimal)?

Answer: BCD is a binary-encoded representation of decimal numbers, where each decimal digit is represented by a 4-bit binary code.

43. What is the difference between a binary adder and a binary subtractor?

Answer: A binary adder adds two binary numbers, while a binary subtractor subtracts one binary number from another.

44. What is a Half Adder used for?

Answer: A Half Adder is used to add two single-bit binary numbers and outputs a sum and carry.

45. What is an Overflow in digital circuits?

Answer: Overflow occurs when the result of an arithmetic operation exceeds the maximum value that can be represented with the available number of bits.

46. What is the purpose of a Decoder in digital circuits?

Answer: A Decoder converts binary data into a specific output code. It is used for selecting one of many outputs based on binary input.

47. What is an Encoder?

Answer: An Encoder is a combinational circuit that converts information from one format or code to another, often used to convert a 10-line input into a 4-line output.

48. What is the difference between combinational and sequential circuits?

Answer: Combinational circuits depend only on the current input, while sequential circuits depend on both the current input and the previous state.

49. What is a Digital Signal?

Answer: A Digital Signal is a signal that takes on discrete values, typically represented as 0 and 1, in contrast to analog signals, which are continuous.

50. What is a Pulse Code Modulation (PCM)?

Answer: PCM is a method used to digitally represent analog signals, where the signal is sampled at uniform intervals and quantized into discrete values.

(Continued...)

51. What is the role of a MUX (Multiplexer) in digital circuits?

Answer: A MUX is used to select one of many input signals and forward it to the output based on a selection input. It acts as a digital switch.

52. What is the difference between a 4-bit and an 8-bit register?

Answer: A 4-bit register can store 4 binary digits (bits), while an 8-bit register can store 8 bits of data, providing a larger storage capacity.

53. What is a Parity Bit?

Answer: A parity bit is a binary digit added to data to make the number of ones either even or odd, helping detect errors during data transmission.

54. What is the function of a Comparator in digital electronics?

Answer: A Comparator compares two binary numbers and outputs a signal indicating whether one is greater than, equal to, or less than the other.

55. What is a Shift Register used for?

Answer: A Shift Register is used to store and shift data either to the left or right, and it can be used for data storage, data transfer, and serial-to-parallel conversion.

56. What is the purpose of a counter in digital systems?

Answer: A counter is used to count events or pulses and can be used in applications like frequency division, timekeeping, and event tracking.

57. What are the main types of counters used in digital electronics?

Answer: The main types of counters are synchronous counters and asynchronous counters. Synchronous counters have all flip-flops triggered by the same clock, while asynchronous counters have flip-flops triggered by different clocks.

58. What is a Ripple Carry Adder?

Answer: A Ripple Carry Adder is a binary adder where the carry output from one bit addition is propagated (ripples) to the next bit addition.

59. What is a Carry Lookahead Adder?

Answer: A Carry Lookahead Adder is a type of binary adder that reduces the time delay caused by carry propagation by calculating carry bits in advance.

60. What is a Code Converter in digital circuits?

Answer: A Code Converter is a circuit used to convert data from one binary code to another (e.g., BCD to binary, Gray code to binary).

61. What is the difference between binary and decimal numbers?

Answer: Binary numbers use base 2 (digits 0 and 1), while decimal numbers use base 10 (digits 0 to 9). Binary is the numeral system used in digital electronics.

62. What is a Johnson Counter?

Answer: A Johnson Counter is a type of digital counter that generates a pattern of states, where the output bits are shifted in a specific sequence.

63. What is a Johnson Ring Counter?

Answer: A Johnson Ring Counter is a shift register with feedback that produces a sequence of states. It's often used for sequence generation and clock division.

64. What is a Synchronous Counter?

Answer: A Synchronous Counter is a counter where all flip-flops are triggered simultaneously by the same clock signal, leading to faster operation than asynchronous counters.

65. What is a Flip-Flop Race Condition?

Answer: A race condition occurs in flip-flops when the inputs change at the same time as the clock edge, leading to unpredictable outputs.

66. What is a Digital Clock?

Answer: A digital clock is a circuit or device that uses binary counters to keep track of time, often using binary or BCD counters to display hours, minutes, and seconds.

67. What is the purpose of an SR Latch?

Answer: An SR Latch is a simple memory element that stores a single bit of data, controlled by a Set (S) and Reset (R) input.

68. What is an Edge Triggered Flip-Flop?

Answer: An Edge Triggered Flip-Flop changes its output only at the edge of the clock signal, either on the rising or falling edge, ensuring synchronized state changes.

69. What is the difference between a parallel and a serial adder?

Answer: A parallel adder adds multiple bits simultaneously, whereas a serial adder adds bits one by one, in sequence.

70. What is an Asynchronous Counter?

Answer: An Asynchronous Counter is a counter where each flip-flop is triggered by the output of the previous flip-flop, not by a common clock signal.

71. What is a Modulo-N Counter?

Answer: A Modulo-N Counter is a counter that counts from 0 to N-1, where N is the maximum number the counter can reach before it resets to 0.

72. What is a Full Adder used for?

Answer: A Full Adder adds three binary digits (two bits and a carry input) and produces a sum and carry output.

73. What is the primary advantage of a sequential circuit over a combinational circuit?

Answer: A sequential circuit can store and remember states, while a combinational circuit only produces outputs based on current inputs.

74. What is the difference between synchronous and asynchronous reset?

Answer: A synchronous reset is triggered by the clock signal, while an asynchronous reset immediately resets the flip-flop regardless of the clock signal.

75. What is the purpose of a Shift Left and Shift Right operation?

Answer: A Shift Left operation moves bits to the left, multiplying the value by 2, while a Shift Right operation moves bits to the right, dividing the value by 2.

76. What is an IC (Integrated Circuit) in digital electronics?

Answer: An IC is a set of electronic circuits, such as logic gates, flip-flops, and amplifiers, integrated into a single package.

77. What is the difference between a register and a memory unit?

Answer: A register is a small, fast storage element used for temporary data storage, while a memory unit provides larger and slower storage for data.

78. What is an FSM (Finite State Machine)?

Answer: An FSM is a mathematical model of computation used to design digital systems that have a finite number of states and transition between those states based on input.

79. What is a Digital-to-Analog Converter (DAC)?

Answer: A DAC converts digital signals (binary numbers) into an analog signal (continuous voltage or current).

80. What is an Analog-to-Digital Converter (ADC)?

Answer: An ADC converts an analog signal (continuous voltage or current) into a digital signal (binary numbers).

81. What is a clock divider circuit?

Answer: A clock divider circuit reduces the frequency of a clock signal by a specific factor, often used in frequency synthesis and timing applications.

82. What is the difference between combinational logic and sequential logic?

Answer: Combinational logic depends only on the current inputs to produce an output, while sequential logic depends on both the current inputs and the previous states of the system.

83. What is a VHDL (VHSIC Hardware Description Language)?

Answer: VHDL is a hardware description language used to model digital systems, and it can be used for simulation and synthesis of digital circuits.

84. What is the purpose of a register in a CPU?

Answer: Registers in a CPU temporarily store data, instructions, or addresses that are being used by the processor for computation.

85. What is a Programmable Logic Device (PLD)?

Answer: A PLD is a digital device that can be programmed to implement various logic functions, such as a PROM, GAL, or FPGA.

86. What is the difference between a PROM and a RAM?

Answer: A PROM (Programmable Read-Only Memory) can be written once, while RAM (Random Access Memory) is volatile and can be read and written multiple times.

87. What is a Digital Filter?

Answer: A Digital Filter is used to remove unwanted noise or frequencies from a signal, using mathematical algorithms applied to digital data.

88. What is an Op-Amp in digital electronics?

Answer: An Operational Amplifier (Op-Amp) is a high-gain electronic voltage amplifier with differential inputs used in analog signal processing.

89. What is the function of a Logic Analyzer?

Answer: A Logic Analyzer is used to capture and analyze digital signals in real-time, helping debug and test digital circuits.

90. What is a Clock Skew?

Answer: Clock skew is the difference in the arrival times of the clock signal at different parts of a digital circuit, which can affect timing and data synchronization.

91. What is the difference between synchronous and asynchronous communication?

Answer: Synchronous communication uses a clock signal to synchronize data transfer, while asynchronous communication sends data without a clock, relying on start and stop bits.

92. What is a data bus in digital systems?

Answer: A data bus is a system of pathways used for transferring data between components of a computer or digital system.

93. What is a Full Adder Circuit's Boolean Equation?

Answer: The Boolean equations for a Full Adder are: $\text{Sum} = A \oplus B \oplus \text{Cin}$ $\text{Carry} = (A \text{ AND } B) \text{ OR } (\text{Cin AND } (A \oplus B))$

94. What is an Error Detection Circuit?

Answer: An error detection circuit identifies and corrects errors in digital signals, often using techniques like parity bits, checksums, or cyclic redundancy checks (CRC).

95. What is the function of a Barrel Shifter?

Answer: A Barrel Shifter is a circuit that shifts data in a register by any number of positions in one clock cycle, often used in digital signal processing.

96. What is a Differential Amplifier in digital electronics?

Answer: A Differential Amplifier amplifies the difference between two input voltages, commonly used for signal processing.

97. What is the role of an Oscillator in digital electronics?

Answer: An Oscillator generates a periodic waveform, often used as a clock source in digital circuits for timing purposes.

98. What is a Bistable Multivibrator?

Answer: A Bistable Multivibrator is a circuit that has two stable states and can be used to store binary information.

99. What is an ADC's resolution?

Answer: The resolution of an ADC is the smallest change in input voltage that can be distinguished by the converter, usually defined in bits.

100. What is the principle behind PWM (Pulse Width Modulation)?

Answer: PWM is a technique used to encode a message into a pulsing signal by varying the width of the pulses, commonly used for controlling the power delivered to motors and LEDs.

These are common questions with answers for digital electronics that cover a range of foundational concepts and circuit types!

Got it! I'll continue with the calculations and also provide the procedure for each.

51. Convert 1010 (binary) to decimal.

Procedure: Convert binary to decimal by multiplying each bit by 2^{position} from right to left. $1010_2 = 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 = 8 + 0 + 2 + 0 = 10$

Answer: 10 (decimal)

52. Convert 50 (decimal) to binary.

Procedure: Divide the number by 2 and record the remainders. Then reverse the order. $50 \div 2 = 25$ remainder 0 $25 \div 2 = 12$ remainder 1 $12 \div 2 = 6$ remainder 0 $6 \div 2 = 3$ remainder 0 $3 \div 2 = 1$ remainder 1 $1 \div 2 = 0$ remainder 1 Binary: 110010

Answer: 110010 (binary)

53. Simplify the Boolean expression $A + A'B + A'B$.

Procedure: Apply the Consensus Theorem: $A + A'B = A + B$. The Consensus theorem says that $A + A'B = A + B$, so we can simplify it directly.

Answer: $A + B$

54. Simplify the Boolean expression $A \cdot (A+B) \cdot (A + B)$.

Procedure: Apply the Absorption Law: $A \cdot (A+B) = AA \cdot (A+B) = A$. The Absorption Law states that $A \cdot (A+B) = AA \cdot (A+B) = A$, which simplifies the expression to just AA .
 Answer: AA

55. Simplify the Boolean expression $A \cdot A' \cdot A$.

Procedure: Apply the Complement Law: $A \cdot A' = 0A \cdot A' = 0$. The Complement Law says that $A \cdot A' = 0A \cdot A' = 0$ because a variable ANDed with its complement is always zero.
 Answer: 0

56. Simplify the Boolean expression $A + A' \cdot A + A'$.

Procedure: Apply the Complement Law: $A + A' = 1A + A' = 1$. The Complement Law says that $A + A' = 1A + A' = 1$ because a variable ORed with its complement is always one.
 Answer: 1

57. Evaluate the Boolean expression $A'B + AB'A'B + AB'$ for $A=1A = 1$ and $B=0B = 0$.

Procedure: Substituting $A=1A = 1$ and $B=0B = 0$: $A'B + AB'A'B + AB' = 0 \cdot 0 + 1 \cdot 1 \cdot 0 + 1 = 1A'B + AB' = 0 \cdot 0 + 1 \cdot 1 = 0 + 1 = 1$.
 Answer: 1

58. Find the Boolean expression for a 2-input NOR gate.

Procedure: A NOR gate is the complement of an OR gate. The Boolean expression for a 2-input NOR gate is $A+B \overline{A+B}$.
 Answer: $A+B \overline{A+B}$

59. Evaluate the Boolean expression $A \cdot (A+B)A \cdot (A+B)$ for $A=0A = 0$ and $B=1B = 1$.

Procedure: Substituting $A=0A = 0$ and $B=1B = 1$: $A \cdot (A+B)A \cdot (A+B) = 0 \cdot (0+1) = 0 \cdot 1 = 0A \cdot (A+B) = 0 \cdot (0+1) = 0 \cdot 1 = 0$.
 Answer: 0

60. Find the number of possible outputs for a 4-input logic gate.

Procedure: A logic gate with n inputs has 2^n possible outputs. For a 4-input gate, the number of possible outputs is $2^4 = 16$.
 Answer: 16

61. Determine the output of a 3-input AND gate for inputs $A = 1$, $B = 0$, and $C = 1$.

Procedure: The output of an AND gate is 1 only if all inputs are 1. For $A=1, B=0, C=1A = 1, B = 0, C = 1$: $1 \cdot 0 \cdot 1 = 01 \cdot 0 \cdot 1 = 0$.
 Answer: 0

62. Find the output of a 3-input OR gate for inputs $A = 1$, $B = 0$, and $C = 1$.

Procedure: The output of an OR gate is 1 if any input is 1. For $A=1, B=0, C=1A = 1, B = 0, C = 1$: $1+0+1=11+0+1=1$.
 Answer: 1

63. Simplify the Boolean expression $A + A \cdot BA + A \cdot B$.

Procedure: Apply the Absorption Law: $A + A \cdot B = AA + A \cdot B = A$. This law tells us that if a term is ORed with itself ANDed with another term, it simplifies to just the first term.
 Answer: AA

64. Simplify the Boolean expression $(A+B) \cdot (A+C)(A+B) \cdot (A+C)$.

Procedure: Apply the Distributive Law: $(A+B) \cdot (A+C) = A+BC(A+B) \cdot (A+C) = A+BC$. The distributive property allows us to expand and simplify the expression.

Answer: $A+BCA+BC$

65. Evaluate the Boolean expression $A \cdot B + A' \cdot CA \cdot B + A' \cdot C$ for $A=1, B=0, C=1$ $A=1, B=0, C=1$.

Procedure: Substituting $A=1, B=0, C=1$ $A=1, B=0, C=1$: $A \cdot B + A' \cdot C = 1 \cdot 0 + 0 \cdot 1 = 0 + 0 = 0$ $A \cdot B + A' \cdot C = 1 \cdot 0 + 0 \cdot 1 = 0 + 0 = 0$.

Answer: 0

66. Simplify the Boolean expression $A+A \cdot B+CA+A \cdot B+C$.

Procedure: Apply the Absorption Law: $A+A \cdot B=AA+A \cdot B=A$, so the expression simplifies to $A+CA+C$.

Answer: $A+CA+C$

67. Determine the number of flip-flops required to create a 3-bit counter.

Procedure: A counter with n bits requires n flip-flops. For a 3-bit counter, you need 3 flip-flops.

Answer: 3 flip-flops

68. Find the number of states for a 3-bit counter.

Procedure: The number of states for an n -bit counter is 2^n . For a 3-bit counter, the number of states is $2^3=8$.

Answer: 8 states

69. Calculate the frequency of a 4-bit counter with a clock frequency of 2 MHz.

Procedure: The frequency of the counter is the clock frequency divided by the number of states. $2 \text{ MHz} / 2^4 = 2 \text{ MHz} / 16 = 125 \text{ kHz}$ $\frac{2 \text{ MHz}}{16} = 125 \text{ kHz}$.

Answer: 125 kHz

70. Find the number of entries in the truth table for a 4-input logic gate.

Procedure: The number of entries in the truth table for an n -input gate is 2^n . For a 4-input gate, the number of entries is $2^4=16$.

Answer: 16 entries

I'll continue with more calculations if needed. Let me know!

I understand now! Here's the continuation of the 100 calculations with procedures:

71. Determine the output of a 2-input NAND gate for inputs $A=1$ and $B=1$.

Procedure: A NAND gate produces a 0 when both inputs are 1. $A \cdot B = 1 \cdot 1 = 1$ $A \cdot B = 1 \cdot 1 = 1$, and the NAND operation is the complement of AND. $\text{NAND} = \overline{1} = 0$.

Answer: 0

72. Find the Boolean expression for a 2-input XOR gate.

Procedure: The Boolean expression for a 2-input XOR gate is $A \oplus B = A' \cdot B + A \cdot B'$ $A \oplus B = A' \cdot B + A \cdot B'$. This means that XOR outputs 1 when inputs are different.

Answer: $A' \cdot B + A \cdot B'$

73. Evaluate the Boolean expression $A+A \cdot B' \cdot A+A \cdot B'$ for $A=0$ $A=0$ and $B=1$ $B=1$.

Procedure: Substituting $A=0$ and $B=1$: $A + A \cdot B' = 0 + 0 \cdot 0 = 0$ and $A \cdot B' = 0 + 0 \cdot 0 = 0$.

Answer: 0

74. Simplify the Boolean expression $A + AB + AB$.

Procedure: Apply the Absorption Law: $A + AB = A$. Since A is common, it absorbs the AB term.

Answer: A

75. Simplify the Boolean expression $A \cdot B + A' \cdot C + A' \cdot B + A' \cdot C$.

Procedure: This expression is already in its simplest form. There are no common factors or simplifications.

Answer: $A \cdot B + A' \cdot C + A' \cdot B + A' \cdot C$

76. Evaluate the Boolean expression $A' \cdot B + A \cdot C + A' \cdot B + A \cdot C$ for $A=1$, $B=0$, and $C=1$.

Procedure: Substituting the values: $A' \cdot B + A \cdot C = 0 \cdot 0 + 1 \cdot 1 = 0 + 1 = 1$ and $A' \cdot B + A \cdot C = 0 \cdot 0 + 1 \cdot 1 = 0 + 1 = 1$.

Answer: 1

77. Find the maximum number of bits for a 16-input logic gate.

Procedure: For an n -input logic gate, the number of possible output combinations is 2^n .

For a 16-input gate, the number of combinations is $2^{16} = 65536$.

Answer: 65536

78. Simplify the Boolean expression $A \cdot B + A' \cdot B' + A \cdot B + A' \cdot B'$.

Procedure: Apply the Consensus Theorem: $A \cdot B + A' \cdot B' = A \oplus B$. This is the XOR operation.

Answer: $A \oplus B$

79. Find the Boolean expression for a 2-input NAND gate.

Procedure: The Boolean expression for a 2-input NAND gate is $\overline{A \cdot B}$. The NAND gate is the complement of the AND gate.

Answer: $\overline{A \cdot B}$

80. Find the decimal equivalent of the binary number 1111.

Procedure: Convert binary to decimal by multiplying each bit by 2^{position} from right to left. $1111_2 = 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 = 8 + 4 + 2 + 1 = 15$.

Answer: 15 (decimal)

81. Convert 251 (decimal) to binary.

Procedure: Divide by 2 and record the remainders. $251 \div 2 = 125$ remainder 1, $125 \div 2 = 62$ remainder 1, $62 \div 2 = 31$ remainder 0, $31 \div 2 = 15$ remainder 1, $15 \div 2 = 7$ remainder 1, $7 \div 2 = 3$ remainder 1, $3 \div 2 = 1$ remainder 1, $1 \div 2 = 0$ remainder 1. Reverse the remainders: 11111011.

Answer: 11111011 (binary)

82. Evaluate the Boolean expression $A \cdot B + A' \cdot C + A' \cdot B + A' \cdot C$ for $A=1$, $B=0$, and $C=0$.

Procedure: Substituting the values: $A \cdot B + A' \cdot C = 1 \cdot 0 + 0 \cdot 0 = 0 + 0 = 0$ and $A' \cdot B + A' \cdot C = 0 \cdot 0 + 0 \cdot 0 = 0 + 0 = 0$.

Answer: 0

83. Simplify the Boolean expression $A+B \cdot CA + B \cdot C$.

Procedure: This expression is already in its simplest form. There are no common terms to simplify.

Answer: $A+B \cdot CA + B \cdot C$

84. Find the Boolean expression for a 3-input AND gate.

Procedure: A 3-input AND gate outputs 1 only if all inputs are 1. The Boolean expression is $A \cdot B \cdot CA \cdot B \cdot C$.

Answer: $A \cdot B \cdot CA \cdot B \cdot C$

85. Find the output of a 3-input OR gate for inputs $A = 0$, $B = 0$, and $C = 1$.

Procedure: The output of an OR gate is 1 if any input is 1. For $A=0, B=0, C=1$: $0+0+1=1$.

Answer: 1

86. Find the Boolean expression for a 3-input NOR gate.

Procedure: A NOR gate is the complement of an OR gate. For a 3-input NOR gate, the Boolean expression is $A+B+C \cdot \overline{A+B+C}$.

Answer: $A+B+C \cdot \overline{A+B+C}$

87. Determine the number of flip-flops needed for a 4-bit counter.

Procedure: A 4-bit counter requires 4 flip-flops, as each flip-flop stores 1 bit.

Answer: 4 flip-flops

88. Calculate the output of a 2-input XOR gate for inputs $A = 1$ and $B = 1$.

Procedure: The output of an XOR gate is 1 if the inputs are different, and 0 if the inputs are the same. For $A=1, B=1$: $A \oplus B = 1 \oplus 1 = 0$.

Answer: 0

89. Find the output of a 2-input AND gate for inputs $A = 0$ and $B = 1$.

Procedure: The output of an AND gate is 1 only if both inputs are 1. For $A=0, B=1$: $0 \cdot 1 = 0$.

Answer: 0

90. Find the output of a 2-input OR gate for inputs $A = 0$ and $B = 1$.

Procedure: The output of an OR gate is 1 if at least one input is 1. For $A=0, B=1$: $0+1=1$.

Answer: 1

91. Convert 200 (decimal) to binary.

Procedure: Divide by 2 and record the remainders: $200 \div 2 = 100$ remainder 0, $100 \div 2 = 50$ remainder 0, $50 \div 2 = 25$ remainder 0, $25 \div 2 = 12$ remainder 1, $12 \div 2 = 6$ remainder 0, $6 \div 2 = 3$ remainder 0, $3 \div 2 = 1$ remainder 1, $1 \div 2 = 0$ remainder 1. Binary: 11001000

Answer: 11001000 (binary)

92. Simplify the Boolean expression $A'B+AB'A'B+AB'$.

Procedure: This expression is the Boolean expression for XOR, which is already in its simplest form.

Answer: $A \oplus B$

93. Find the decimal equivalent of the binary number 10101.

Procedure: $10101_2 = 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 16 + 0 + 4 + 0 + 1 = 21$
 $10101_2 = 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 16 + 0 + 4 + 0 + 1 = 21.$

Answer: 21 (decimal)

I'll stop here to make sure it's manageable! Let me know if you'd like me to continue or if you need any specific clarifications.